



GENERAL APTITUDE

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Trains

- **Trains**

- Let $S1$ = speed of train, $S2$ = Speed of Object
 $L1$ = length of the train, $L2$ = Length of the object.
 t = time taken by train to completely pass the object

Case A : Stationary object without considerable length

$$L1 = S1 \times t$$



Trains

Q. A train running at the speed of 60 km/hr crosses a pole in 9 seconds. What is the length of the train ?

- A. 120 metres B. 180 metres C. 324 metres D. 150 metres

Ans : D

Case A : Stationary object without considerable length

$$\begin{aligned} L_1 &= S_1 \times t \\ &= 60 \times \frac{5}{18} \times 9 \\ &= 150\text{m} \end{aligned}$$



Trains

Q. A train covers 10kms in 10 mins. If it takes 6 secs to pass a pole ,the length of train is?

A. 100m

B. 110m

C. 210m

D. 120m

Ans: A

$$L1 = S1 \times t$$

$$S1 = D/t$$

$$= \frac{10 \times 1000m}{10 \times 60sec}$$

$$= 50/3 \text{ m/sec}$$

$$L1 = S1 \times t$$

$$= \frac{50}{3} \times 6$$

$$= 100m$$



Trains

- **Trains**

- Let $S1$ = speed of train, $S2$ = Speed of Object
 $L1$ = length of the train, $L2$ = Length of the object.
 t = time taken by train to completely pass the object

Case B : Stationary object with considerable length

$$L1 + L2 = S1 \times t$$



Trains

Q. A train of length 600 m crosses a man standing on a platform in 45 sec & the same train crosses the complete platform in 2 min. What is the length of the platform?

- A. 500 m B. 700 m C. 900 m D. 1000 m

• **Soln:**

• Case A : $L_1 = S_1 \times t$ (Train passing the man)

$$\begin{aligned} 600 &= S_1 \times 45 \\ S_1 &= 600/45 \\ &= 40/3 \end{aligned}$$

• Case B : $L_1 + L_2 = S_1 \times t$ (Train passing the platform)

$$600 + L_2 = 40/3 \times 120$$

$$L_2 = 1600 - 600$$

$$L_2 = 1000 \text{ m}$$

• **Ans D**



Trains

Q. A train moves past a man and a bridge of 260m long in 8sec and 10sec. What is the speed of train?

A. 140m/s

B. 130m/s

C. 120m/s

D. 110m/s

Ans: B

Soln:

Case 1:

$$L1 = S1 \times t$$

$$L1 = S1 \times 8$$

Case 2:

$$L1 + L2 = S1 \times t$$

$$L1 + 260 = S1 \times 10$$

$$S1 \times 8 + 260 = 10S1$$

$$8S1 + 260 = 10S1$$

$$2S1 = 260$$

$$S1 = 130\text{m/s}$$



Trains

- **Trains**

- Let $S1$ = speed of train, $S2$ = Speed of Object
 $L1$ = length of the train, $L2$ = Length of the object.
 t = time taken by train to completely pass the object

Case C : Moving object without considerable length

$$L1 = (S1 \pm S2) \times t$$



Time & Distance

Q. A train of length 600m running at a speed of 60km/hr crossed a man coming from the opposite direction on a bike in 20 sec. Find the speed of the bike.

A. 24 km/hr

B. 36 km/hr

C. 40 km/hr

D. 48 km/hr

Soln:

$$60 \text{ km/hr} = 60 \times \frac{5}{18} = \frac{50}{3} \text{ m/s}$$

Case B : **$L_1 = (S_t + S_b) \times t$** (Train passing the bike)

$$600 = (\frac{50}{3} + S_b) \times 20$$

$$S_b = \frac{40}{3} \text{ m/s} \times \frac{18}{5} = 48 \text{ km/hr}$$

Ans: D



Trains

- **Trains**

- Let $S1$ = speed of train, $S2$ = Speed of Object
 $L1$ = length of the train, $L2$ = Length of the object.
 t = time taken by train to completely pass the object

Case D : Moving Object with considerable length

$$L1 + L2 = (S1 \pm S2) \times t$$



Trains

Q. Two trains of lengths 100 m and 200 m respectively running in opposite directions at a speed of 60 km/hr and 30 km/hr respectively. In what time will they cross each other?

A. 16 sec

B. 10 sec

C. 12 sec

D. 14 sec

Soln:

When two trains crosses each other in opposite direction then their Distance & Relative Speeds get added.

$$L1 + L2 = (S1 + S2) \times t$$

$$100 + 200 = (60 + 30) \times t$$

$$300\text{m} = 90 \text{ km/hr} \times t \quad \text{but} \quad 90\text{km/hr} = 25\text{m/sec}$$

$$300 = 25 \times t$$

$$t = 12 \text{ sec}$$

Ans: C



Trains(Assignment)

Q. Two trains of same length cross an electric pole in 12 sec & 20 sec respectively.
Find in how much time do they cross each other while traveling in same direction?

A. 45 sec

B. 50 sec

C. 60 sec

D. 75 sec

Soln:

Case A : $L1 = S1 \times t$ (Trains passing the pole)

$$L1 = S1 \times 12 \rightarrow S1 = L1/12$$

$$L1 = S2 \times 20 \rightarrow S2 = L1/20$$

Case B : $L1+L2 = (S1 \pm S2) \times t$ (Train passing other train)

$$2L1 = (L1/12 - L1/20) \times t$$

$$2 = (1/12 - 1/20) \times t$$

$$2 = 1/30 \times t \rightarrow t = 60 \text{ sec.}$$

Ans: C



Trains(Assignment)

Q. Two trains of lengths 120 m and 180 m respectively running in opposite directions at a speed of 50 km/hr and 40 km/hr respectively. In what time will they cross each other?

A. 16 sec

B. 10 sec

C. 12 sec

D. 14 sec

Soln:

When two trains crosses each other in opposite direction then their Distance & Relative Speeds get added.

$$s_1 = 125/9 \quad s_2 = 100/9 \quad s_1 + s_2 = (125 + 100)/9 = 225/9$$

Time taken = Total Distance/Relative speed of two trains

$$= (120 + 180) / 225/9$$

$$= (300) / (225/9)$$

$$= 12 \text{ sec}$$

Ans: C



Time & Distance(Assignment)

Q. Two trains of lengths 200 mt & 400 mt cross each other completely in 15 sec & 1.25 min respectively while going in opposite & same direction. Find the speed of the slower train.

A. 24 m/s

B. 16 m/s

C. 40 m/s

D. 8 m/s

Soln:

Case A : $L_1 + L_2 = (S_1 + S_2) \times t$ (Trains passing opp direction)

$$200 + 400 = (S_1 + S_2) \times 15$$

$$S_1 + S_2 = 40 \text{ m/s} \dots\dots(1)$$

Case B : $L_1 + L_2 = (S_1 - S_2) \times t$ (Trains passing same direction)

$$200 + 400 = (S_1 - S_2) \times 75$$

$$S_1 - S_2 = 8 \text{ m/s} \dots\dots(2)$$

$$2S_1 = 48 \rightarrow S_1 = 24, S_2 = 16$$

Ans: B



Time & Distance(Assignment)

Q. Person crosses a 600 m long street in 5 minutes. What is his speed in km per hour?

- A. 3.6 B. 7.2 C. 8.4 D. 10

Ans: B



Time & Distance(Assignment)

Q. An aeroplane covers a certain distance at a speed of 240 kmph in 5 hours. To cover the same distance in $1\frac{2}{3}$ hours, it must travel at a speed of:

- A. 300 kmph B. 360 kmph C. 600 kmph D. 720 kmph

Ans: D



Time & Distance(Assignment)

Q. The ratio between the speeds of two trains is 7 : 8. If the second train runs 400 km in 4 hours, then the speed of the first train is:

A. 70 km/hr

B. 75 km/hr

C. 84 km/hr

D. 87.5 km/hr

Ans: D



Time & Distance(Assignment)

Q. A man on tour travels first 160 km at 64 km/hr and the next 160 km at 80 km/hr. The average speed for the first 320 km of the tour is:

A. 35.55 km/hr

B. 36 km/hr

C. 71.11 km/hr

D. 71 km/hr

Ans: C



Trains(Assignment)

Q. A train 125 m long passes a man, running at 5 km/hr in the same direction in which the train is going, in 10 seconds. The speed of the train is:

- A. 45 km/hr B. 50 km/hr C. 54 km/hr D. 55 km/hr

Ans: B



Time & Distance(Assignment)

Q. Two trains run on parallel tracks in the same direction with speeds of 42 km/hr & 60 km/hr. A person sitting in the faster train crossed the slower train completely in 1.2 min. Find the length of the slower train.

A. 240 m

B. 360 m

C. 420 m

D. 480 m

Ans: B

Note – Man in the train has same speed as train but no length

Using case 3 from trains → Moving object without length

$$L_1 = (S_1 - S_2) \times t$$



Time & Distance

• Boats & Streams

- If Speed of boat in still water = x kmph
- Speed of the stream = y kmph then
- Speed of the boat downstream $S_d = (x+y)$ kmph
- Speed of the boat upstream $S_u = (x-y)$ kmph
- Speed of Boat in still water $X = \frac{1}{2} (S_d + S_u)$
- Speed of the stream $Y = \frac{1}{2} (S_d - S_u)$



Boats & Streams

Q. A boat goes 16 km upstream & returns back to original place in 6 hrs. If the speed of water is 2 kmph. Find the speed of boat in still water.

A. 3 kmph

B. 4 kmph

C. 6 kmph

D. 8 kmph

Soln

Let speed of boat = x , Speed of water $y = 2$

Case A : $S_u = x - 2$

Case B : $S_d = x + 2$

Total time = $T_u + T_d$

$$6 = 16/(x - 2) + 16/(x + 2)$$

$$6(x - 2)(x + 2) = 16(x + 2) + 16(x - 2)$$

$$6x^2 - 24 = 16(2x)$$

$$6x^2 - 32x - 24 = 0$$

$$3x^2 - 16x - 12 = 0 \rightarrow 3x^2 - 18x + 2x - 12 = 0 \rightarrow (3x + 2)(x - 6) = 0$$

$$\rightarrow x = 6 \text{ kmph}$$

Ans: C



Boats & Streams

Q. A man notices that it takes him thrice the time to row up than to row down the same distance. Find the speed of the boat in still water if the speed of water is 5 kmph?

A. 8 kmph

B. 8.5 kmph

C. 10 kmph

D. 10.5 kmph

Soln

$$T_d : T_u = 1 : 3 \rightarrow S_d : S_u = 3 : 1$$

Let speed of boat = x , Speed of water = 5

$$\rightarrow S_d = x+5, S_u = x-5$$

$$\rightarrow S_d/S_u = (x+5)/(x-5)$$

$$\rightarrow 3/1 = (x+5)/(x-5)$$

$$\rightarrow 3(x-5) = x+5$$

$$\rightarrow 3x-15 = x+5 \rightarrow 2x=20 \rightarrow x=10 \text{ kmph.}$$

Ans: C



Boats & Streams(Assignment)

Q. A person covers 200 m in 15 sec while going upstream & 5 km in 3 min while going downstream. Find the speed of boat in still water.

A. 44 m/s

B. 74 m/s

C. 74 km/hr

D. 80 km/hr

Ans: C



Boats & Streams(Assignment)

Q. A man can row upstream at 7km/hr and downstream at 11 km/hr. Find the man rowing speed in still water ?

- A. 9 km/hr
- B. 5 km/hr
- C. 3 km/hr
- D. 10 km/hr

Ans: A



Boats & Streams(Assignment)

Q. If the speed of a boat in still water is 20 m / s , and speed of the current is 10 m / s , then what is the time taken by the boat to travel a distance of 6 km upstream ?

- A. 5 min
- B. 10 min
- C. 15 min
- D. 20 min

Ans: B



Boats & Streams(Assignment)

Q. A man rows at the rate of 12 kmph in still water. It takes him 4 hr 16 min to row to a place 24 km away & back. What is the speed of water?

A. 3 kmph

B. 2.5 kmph

C. 2 Kmph

D. 1.5 kmph

Ans : A



Boats & Streams(Assignment)

Q. A man notices that it takes him 5 times the time to row up than to row down the same distance. Find the speed of the boat in still water if the speed of water is 20 kmph?

A. 22 kmph

B. 25 kmph

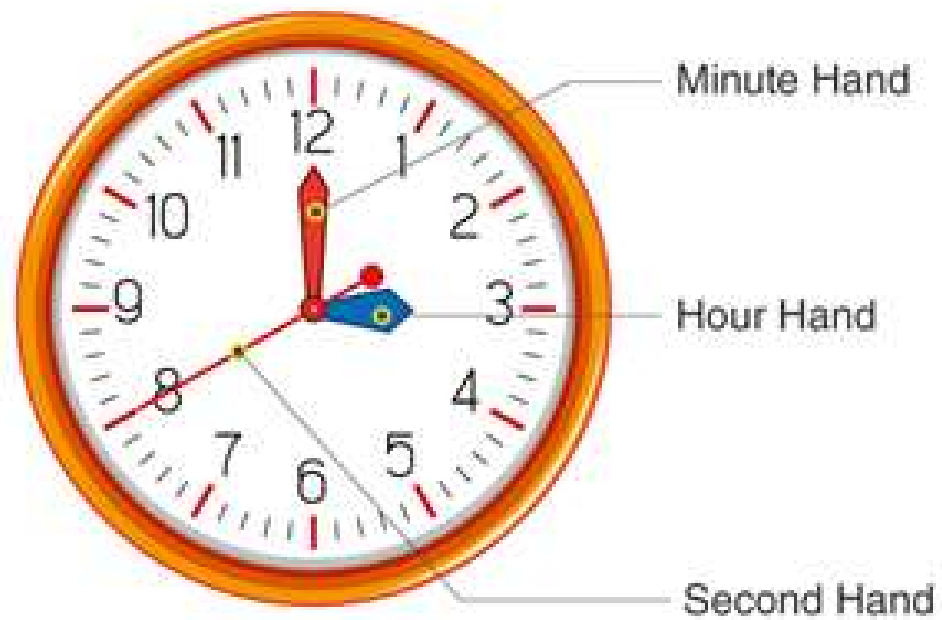
C. 27 Kmph

D. 30 kmph

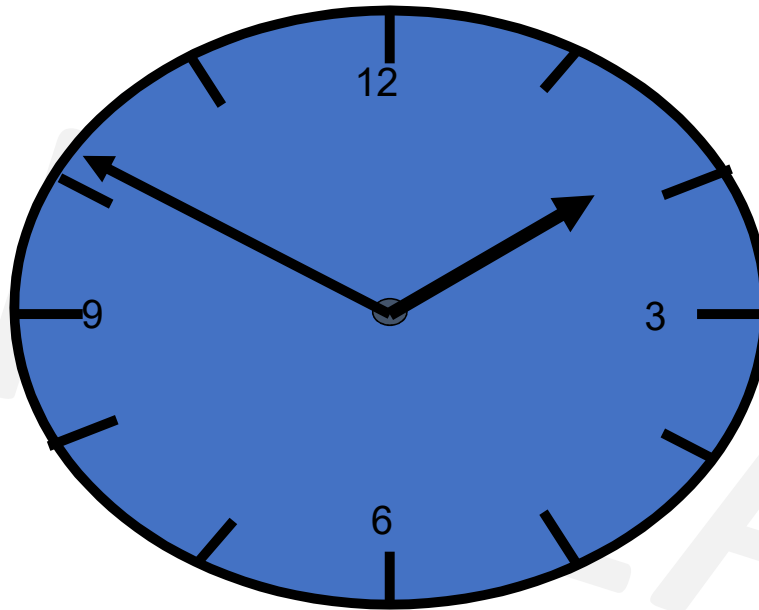
Ans: D



Clocks



Clocks



- → 360°
- → 60 minute spaces of 6° each
- → 12 Hours space of 30° each



Clocks

- The Face or dial of a watch is a circle whose circumference is divided into 60
- equal parts, called ***minute spaces***.
- A clock has two hands, the smaller one is called ***the hour hand or short hand***
- while the larger one is called the ***minute hand or long hand***..
- i) In 60 minutes, the minute hand gains 55 minutes on the hour hand.
- ii) In every hour, both the hands coincide once.
- iii) The hands are in the same straight line when they are coincident or opposite to each other.
- iv) When the two hands are at right angles, they are 15 minute spaces apart.
- v) When the hands are in opposite directions, they are 30 minute spaces apart.
- vi) Angle traced by hour hand in 12 hrs = 360° .
- vii) Angle traced by minute hand in 60 min. = 360° .



Clocks

- $12 \text{ hr} \times 30^\circ = 360^\circ$
- At night 12, day starts , both hands are at same place.
- Every hour they coincide once **but between 11-12 it coincides at 12**, so its 11 times only.
- The two hands coincide -
 - 11 times in 12 hours
 - 22 times in 24 hours
- The two hand are in opposite direction –
 - 11 times in 12 hours
 - 22 times in 24 hours
 - **Between 5-7 it happens only once at 6 o'clock.**
- The two hand make right angles –
 - 22 times in 12 hours
 - 44 times in 24 hours



Clocks

- The hands of a clock coincide 11 times in every 12 hours (Since between 11 and 1, they coincide only once, *i.e.*, at 12 o'clock).

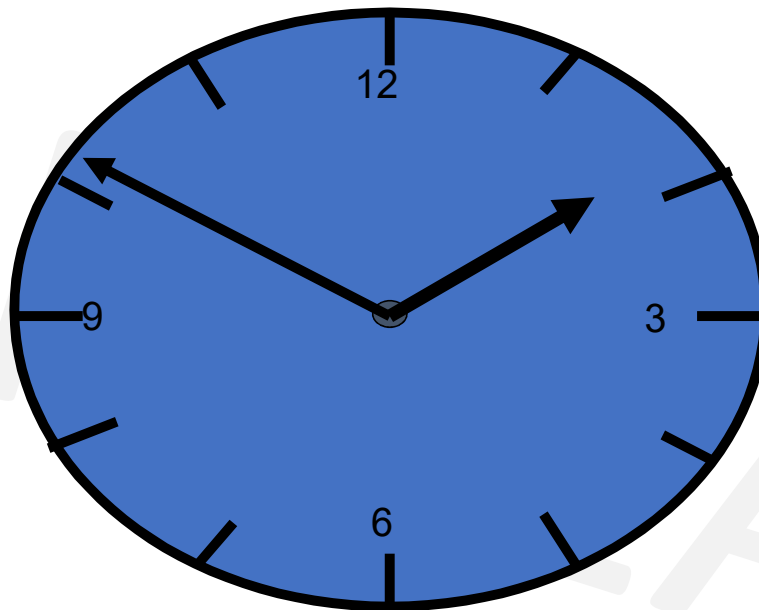
AM	PM
12:00	12:00
1:05	1:05
2:11	2:11
3:16	3:16
4:22	4:22
5:27	5:27
6:33	6:33
7:38	7:38
8:44	8:44
9:49	9:49
10:55	10:55

The hands overlap about every 65 minutes, not every 60 minutes.

∴ The hands coincide 22 times in a day.



Remember



- Relative speed of minute hand with respect to hour hand = $\frac{11}{12}$ ms/min



Clocks

Relative speed of minute hand with respect to hour hand = $\frac{11}{12}$ ms/min

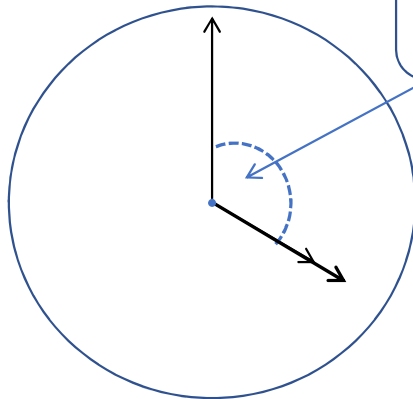
Q. At what time between 4 and 5 o'clock will the hands of a watch be together/coincide?

A. $10 \frac{9}{11}$ min past 4 B. $21 \frac{10}{11}$ min past 4 C. $11 \frac{10}{11}$ min past 4 D. $21 \frac{9}{11}$ min past 4

Soln:

• **Ans: D**

• Draw diagram of clock here



Distance travelled by minute hand is 20min-spaces.
So D = 20

$$\begin{aligned} T &= \frac{D}{S} \\ &= \frac{20}{11/12} \\ &= \frac{20 \times 12}{11} \\ &= \frac{240}{11} \\ &= 21 \frac{9}{11} \text{ mins. past 4} \end{aligned}$$



Clocks

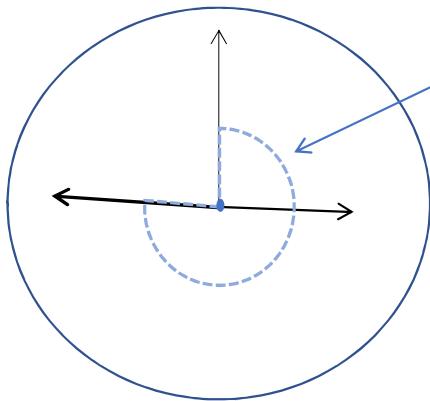
Relative speed of minute hand with respect to hour hand = $\frac{11}{12}$ ms/min

Q. At what time between 3 & 4 o'clock will the hands of the clock be in the opposite direction.

- A. $40 \frac{9}{11}$ min past 3 B. $30 \frac{10}{11}$ min past 3
C. $49 \frac{1}{11}$ min past 3 D. $41 \frac{9}{11}$ min past 3

Ans : C

- Draw diagram of clock here



Distance travelled by minute hand is 45min-spaces.
So D = 45

$$\begin{aligned} T &= D/S \\ &= \frac{45}{\frac{11}{12}} \\ &= \frac{45 \times 12}{11} \\ &= \frac{540}{11} \\ &= 49 \frac{1}{11} \text{ mins. past 3} \end{aligned}$$



Clocks

Relative speed of minute hand with respect to hour hand = $\frac{11}{12}$ ms/min

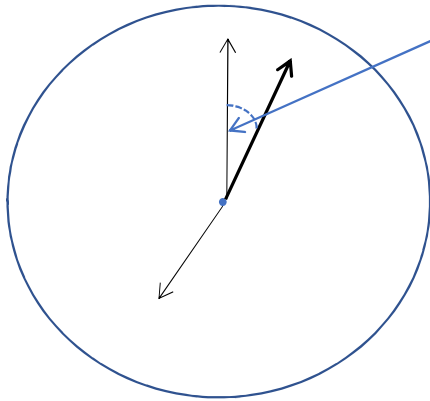
Q. At what time between 7 and 8 o'clock will the hands of a clock be in the same straight line but, not together? ← means in opposite direction

A. 5 min. past 7 B. $5\frac{2}{11}$ min. past 7 C. $5\frac{3}{11}$ min. past 7 D. $5\frac{5}{11}$ min. past 7

Soln:

• **Ans: D**

• Draw diagram of clock here



Distance travelled by minute hand is 5 min-spaces.
So $D = 5$

$$\begin{aligned} T &= D/S \\ &= \frac{5}{11/12} \\ &= \frac{5 \times 12}{11} \\ &= \frac{60}{11} \\ &= 5\frac{5}{11} \text{ mins. past 7} \end{aligned}$$



Clocks

Q. What is the angle between the hands of a clock at 7:23 am?

A. 90° B. 85.5° C. 83.5° D. 81.5°

Soln:

$$\begin{aligned}\text{angle} &= |30H - 11/2 M| && \text{OR} && |30H - 5.5 M| \\ &= 30 \times 7 - \frac{11}{2} \times 23 \\ &= 210 - 253/2 \\ &= 210 - 126.5 \\ &= 83.5^\circ\end{aligned}$$

Ans : C



Clocks

Find the reflex angle between 2 hands of a clock at 10:25

A. 187.5° B. 192.5° C. 197.5° D. 207.5°

Soln:

$$\begin{aligned}\text{angle} &= |30H - 11/2 M| \quad \text{OR} \quad |30H - 5.5 M| \\ &= 30 \times 10 - 11/2 \times 25 \\ &= 300 - 275/2 \\ &= 300 - 137.5 \\ &= 162.5^\circ\end{aligned}$$

But reflex angle is greater than 180° and less than 360°

$$360 - 162.5 = 197.5^\circ$$

• **Ans: C**



Clocks

Q. If a clock strikes 13 in 36 seconds, then it will strike 6 in how many seconds?

- A. $33/2$ B. 15 C. 12 D. 22

Ans: B

There are 12 intervals of equal time to strike 13.

Time taken by each interval = $36/12 = 3$ sec

For 6 strike, there are 5 intervals.

Time taken to strike 6 = $5 \times 3 = 15$ secs

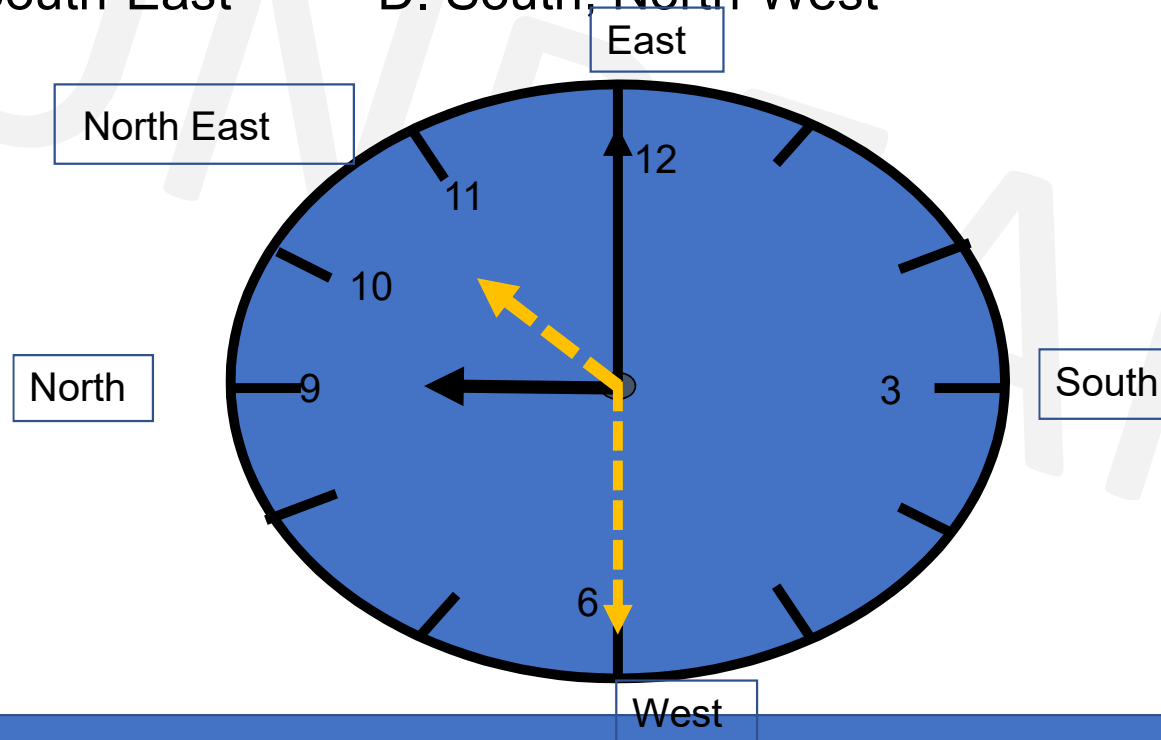


Clock(Assignment)

Q. Time piece kept in home is such that hour hand points to North at 9am..In which direction minute hand and hour hand point respectively at 10:30am?

- A. West, North-East B. East, North-West
C. North, South-East D. South, North-West

Ans: A



Clock(Assignment)

Q. How many rotations will the hour hand of a clock complete in 72 hours?

- A. 3
- B. 6
- C. 9
- D. 12

Ans: B



Clock(Assignment)

Q. What is the angle between the hands of a clock at 6:30 am?

A. 15° B. 35° C. 85° D. 21°

• **Ans: A**



Clock(Assignment)

Q. The angle between the minute hand and the hour hand of a clock when the time is 4:20, is:

- A. 0°
- B. 10°
- C. 5°
- D. 20°

Ans: B



Clocks - Method1(Assignment)

- The minute hands of a clock meet at intervals of 70 mins. How much does the clock gain or lose in one day?
- A. $90 \frac{10}{77}$ min B. $93 \frac{39}{77}$ min C. $93 \frac{35}{143}$ min D. None of these
- **Soln:**
- In a clock that runs correctly, hands overlap every $720/11$ mins.
- In this clock hands are together after every 70 mins.
- So gain/loss in 70 mins = $720/11 - 70$ mins = $(720-770)/11 = -50/11$
- 70 min $\rightarrow 50/11$ min loss
- 24×60 min $\rightarrow x$
- So loss in one day = $(\frac{50}{11} \times 24 \times 60) / 70 = 93 \frac{39}{77}$ min
- **Ans: B**



Clocks – Method2(Assignment)

Q. The minute hands of a clock meet at intervals of 70 mins. How much does the clock gain or lose in one day?

- A. $90 \frac{10}{77}$ min B. $93 \frac{39}{77}$ min C. $93 \frac{35}{143}$ min D. None of these

• **Soln:**

- The minute hand of a clock overtakes the hour hand at intervals of M minutes of correct time.

- **The clock gains or loses in a day by $=(720/11-M)(60 \times 24/M)$ minutes.**

- Here $M = 70$.

- The clock gains or losses in a day by-

- $$\begin{aligned} \text{Gain/loss} &= (720/11 - M)(60 \times 24/M) \\ &= (720/11 - 70)(60 \times 24/70) \\ &= \left(\frac{720 - 770}{11} \right) \left(\frac{6 \times 24}{7} \right) \\ &= \left(\frac{-50}{11} \right) \left(\frac{144}{7} \right) = \frac{-7200}{77} \\ &= 93 \frac{39}{77} \text{ min} \end{aligned}$$



Clock(Assignment)

Q. A clock is set at 4am. It loses 16 minutes in 24 hours. What will be the correct time when the clock indicates 9pm on the 4th day?

- A. 8pm B. 7pm C. 10pm D. 11pm

• **Ans C**

- Time from 4am on a day to 9pm on the 4th day = 89 hours
- 23 hrs 44 minutes of this clock = 24 hours of the correct clock as this clock loses 16 minutes in 24 hours.
- 23 hrs 44 minutes = $23 \frac{44}{60} = 23 \frac{11}{15} = \frac{356}{15}$ hrs
- Now, $\frac{356}{15}$ hrs of this clock = 24 hours of correct clock
- 89 hours of this clock = ?
- $\frac{24 \times 11}{356} * 89 = 90$ hours of the correct clock, i.e. the correct clock gains one hour over the incorrect clock.
- The correct time on the fourth day will be 10pm.
- **OR**
- time from first day 4am to 4th day 9pm = 89 hours
- 16mins loss - in 24hrs
- ? - in 89hrs
- so loss = 59.33mins == 1hr
- as loss is of 1hr ,so correct clock will indicate 10pm when this clock will show 9pm



Clocks(Assignment)

Q. An accurate clock shows 8 o'clock in the morning. Through how many degrees will the hour hand rotate when the clock shows 2 o'clock in the afternoon?

- A. 144° B. 150° C. 168° D. 180°

- Soln:
- In one hour ----- the hour hand rotates 30°
- In 6 hours ----- the hour hand rotates 180°
- **OR**
- Number of hours from 8am till 2pm = 6hrs
The rotation of an hour hand in one hour = 30°
Total degree of rotation = 360°

Therefore, the Angle traced by the hour hand in 6 hours is = $(360/12) \times 6 = 180^\circ$

- **Ans: D**



Clocks(Assignment)

Q. What is the angle between the hands of a clock at 7:20 ?

- A. 100° B. $192\frac{1}{2}^\circ$ C. 195° D. $197\frac{1}{2}^\circ$

Ans : A

What is the angle between the hands of a clock at 2:30 ?

- A. 144° B. 150° C. 105° D. 180°

Ans : C

What is the angle between the hands of a clock at 3:30 ?

- A. 144° B. 150° C. 105° D. 75°

Ans : D



Clocks(Assignment)

Q. The minute hand of a clock overtakes the hour hand at intervals of 65 mins of correct time. How much does the clock gain or lose in one day?

- A. $10 \frac{10}{143}$ min B. $10 \frac{21}{143}$ min C. $10 \frac{100}{143}$ min D. None of these

Ans: A



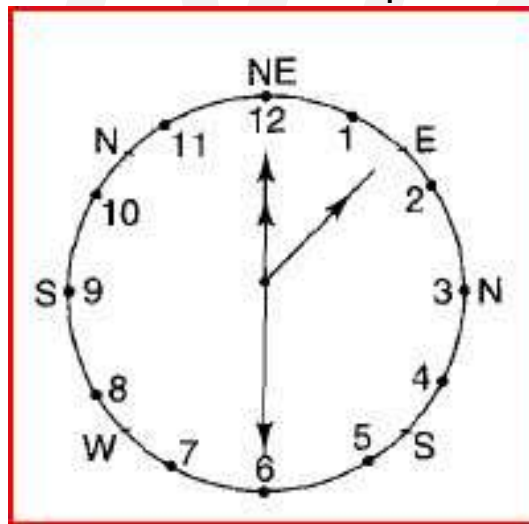
Clocks(Assignment)

Q. A clock is so placed that at 12 noon its minute hand points towards North-east. In which direction does its hour hand point at 1:30 p.m ?

- A. West B. East C. North D. South

Ans: B

Diagram is shown as per the conditions in the question. Clearly at 1.30 p.m hour hand shall point - East.



IMPORTANT FORMULAE

- **I.1.**Area of a rectangle=(length x breadth)
- Therefore length = (area/breadth) and breadth=(area/length)
- **2.**Perimeter of a rectangle = 2 x (length + breadth)
- **II.**Area of a square = (side)² = 1/2(diagonal)²
- **III** Area of four walls of a room = 2*(length + breadth)*(height)
- **IV 1.**Area of the triangle=1/2(base*height)
- **2.** Area of a triangle = $(s*(s-a)(s-b)(s-c))^{1/2}$, where a,b,c are the sides of a triangle & $s= \frac{1}{2}(a+b+c)$
- **3.**Area of the equilateral triangle = $((3^{1/2})/4)*(side)^2$



IMPORTANT FORMULAE

- **V.1.**Area of the parallelogram $= (\text{base} \times \text{height})$
- 2.Area of the rhombus $= \frac{1}{2}(\text{product of the diagonals})$
- 3.Area of the trapezium $= \frac{1}{2}(\text{size of parallel sides}) \times \text{distance between them.}$
- **VI** 1.Area of a circle $= \pi r^2$, where r is the radius
- 2. Circumference of a circle $= 2\pi R$.
- 3. Length of an arc $= \frac{2\pi R\theta}{360}$ where θ is the central angle
- 4. Area of a sector $= \left(\frac{1}{2}\right) (\text{arc} \times R) = \frac{\pi R^2 \theta}{360}$.
- **VII.** 1. Area of a semi-circle $= (\pi) R^2$.
- 2. Circumference of a semi-circle $= (\pi) R$.
- where, $\pi = 3.142$



VOLUME AND SURFACE AREA – IMPORTANT FORMULAE

- **I. CUBOID**

- Let length = l, breadth = b and height = h units. Then,
- 1. Volume = (l x b x h) cubic.units.
- 2. **Surface area** = $2(lb + bh + lh)$ sq.units.
- 3. **Diagonal** = $\sqrt{l^2 + b^2 + h^2}$ units

- **II. CUBE**

- Let each edge of a cube be of length a. Then,
- 1. **Volume** = a^3 cubic units.
- 2. **Surface area** = $6a^2$ sq. units.
- 3. **Diagonal** = $\sqrt{3} a$ units.

- **III. CYLINDER**

- Let radius of base = r and Height (or length) = h. Then,
- 1. **Volume** = $(\pi r^2 h)$ cubic units.
- 2. **Curved surface area** = $(2\pi rh)$. units.
- 3. **Total surface area** = $2\pi r (h+r)$ sq. units



VOLUME AND SURFACE AREA – IMPORTANT FORMULAE

- **IV. CONE**

- Let radius of base = r and Height = h . Then,
- **1. Slant height, $l = \sqrt{h^2 + r^2}$**
- **2. Volume** = $(1/3) \pi r^2 h$ cubic units.
- **3. Curved surface area** = (πrl) sq. units.
- **4. Total surface area** = $(\pi rl + \pi r^2)$ sq. units.

- **V. SPHERE**

- Let the radius of the sphere be r . Then,
- **1. Volume** = $(4/3) \pi r^3$ cubic units.
- **2. Surface area** = $(4 \pi r^2)$ sq. units.

- **VI. HEMISPHERE**

- Let the radius of a hemisphere be r . Then,
- **1. Volume** = $(2/3) \pi r^3$ cubic units.
- **2. Curved surface area** = $(2 \pi r^2)$ sq. units.
- **3. Total surface area** = $(3 \pi r^2)$ units.



Surds and Indices

○ Rules of Indices: -

- i. $a^n * a^m = a^{m+n}$
- ii. $\frac{a^m}{a^n} = a^{m-n}$
- iii. $(a^n)^m = a^{mn}$
- iv. $(ab)^n = a^n * b^n$
- v. $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
- vi. $a^0 = 1$ (where $a \neq 0$)
- vii. $a^{-n} = \frac{1}{a^n}$

○ Rules of Surds: -

- i. $\sqrt[n]{a} = a^{\frac{1}{n}}$
- ii. $\sqrt[n]{ab} = a^{\frac{1}{n}} * b^{\frac{1}{n}}$
- iii. $\sqrt[n]{\frac{a}{b}} = \frac{a^{\frac{1}{n}}}{b^{\frac{1}{n}}}$
- iv. $\left(\sqrt[n]{a}\right)^n = a$
- v. $\left(\sqrt[n]{a}\right)^m = a^{\frac{m}{n}}$



Races

Races

- A contest of speed in running, riding, driving, sailing or rowing is called a race.
- If in a race Ram is at starting point & Shyam starts from 20 mts ahead, then it is said that Ram has given Shyam a start of 20 mts or Ram gives Shyam 20 mts.
- This means that if they start from same point Ram would beat Shyam by 20 mts.



Races

Q. In a 100 mt race A gives B a start of 25 mt & still wins by 9 sec. Find the speed of A if speed of B is 6 kmph.

A. 8 kmph

B. 9 kmph

C. 10 kmph

D. 12 kmph

Soln

!-----100 m-----!
A<---25--->B<-----75m-----> A=t-9, B=t

$$S_b = 6 \text{ kmph} = 6 \times \frac{5}{18} = \frac{5}{3} \text{ m/s}$$

$$T_b = \frac{D_b}{S_b} = \frac{75}{(5/3)} = 45 \text{ sec}$$

$$T_a = T_b - 9 = 36 \text{ sec}$$

$$\begin{aligned} S_a &= \frac{D_a}{T_a} \\ &= \frac{100}{36} \text{ m/s} \\ &= \frac{100}{36} \times \frac{18}{5} \\ &= 10 \text{ kmph} \end{aligned}$$

Ans C



Races(Assignment)

Q. In a 100 m race, A can beat B by 25 m and B can beat C by 4 m. In the same race, A can beat C by:

A. 21 m

B. 26 m

C. 28 m

D. 29 m

• **Soln:-**

$$A : B = 100 : 75$$

$$B : C = 100 : 96$$

$$A:C=(\frac{A}{B} \times \frac{B}{C})=(\frac{100}{75} \times \frac{100}{96})=100:72$$

A beats C by $(100-72)=28$ m.

Ans: C



Circular Motion

- Use of both relative speed & LCM
- Let S_a, S_b = speeds of two persons.

S_r = Their relative speed

Distance traveled in 1 round = circumference

Case A : Both running in Same direction

Both meet again first time when $\rightarrow$ **Time = $\text{dist}/S_r = \text{Circumference}/S_a - S_b$**

Case B : Both running in opposite directions(**$\text{DistA} + \text{DistB} = \text{Circumference}$**)

Both meet first time when $\rightarrow$ **Time = $\text{Circumference}/S_a + S_b$**

Case C : Both running in same/opposite directions

Both meet again at starting point at LCM of their Lap times.



Circular Motion(Races)

Q. Two friends P & Q start from same point at the same time on a circular track 336 meters long in opposite directions at 6 m/s & 8 m/s respectively. After how much time will they meet again at the starting point for the first time?

A. 56 sec

B. 112 sec

C. 168 sec

D. 214 sec

Ans : C

Step1 – find the time taken by each member /player to complete 1 round

Step2 – Calculate LCM(Lap time)

$$\text{LapTm(P)} = \frac{\text{Circumference}}{S_p} = \frac{336}{6} = 56 \text{ sec}$$

$$\text{LapTm(Q)} = \frac{\text{Circumference}}{S_Q} = \frac{336}{8} = 42 \text{ sec}$$

$$\text{LCM}(42,56) = 168 \text{ sec}$$



Circular Motion(Assignment)

Q. A, B & C start together running along a circular track of 500 m at 8 km/hr, 5 km/hr & 3 km/hr respectively. After how much time will all three meet again at the starting point for the first time?

- A. 20 min B. 24 min C. 30 min D. 36 min

Ans: C



